MLOps ke interview me bhi terraform, pipeline, azure hi poochega...

Daily Diary is very important

First Page

What?- Paisa kamana hai

Why? = Shadi karna hai.. km package me na gf mil rahi hai na biwi..

Biwi ko honeymoon pr europe leke jaenge..

How? = DevOps ka rasta bdia lg raha hai.. Baki bhi logo ka hua hai toh mera bhi ho jaega..

Second Page

1. Plan
   - Pehle Terraform Dhua Dhua karna hai
   - Pipeline + Azure Cloud
2. **Story Telling** for Project Use case
   - Industrial IOT, Hospital.. Esi esi industry ke use case banao jo duniya me kisi ne suna hi nahi ho..
     Banking ka basic, todo app.. ye kuch bolna hi nahi hai..

DevOps ki job se paise kamao..

↓

Vo paise se ghoomne jao

↓

Ghoomne ke Vlog bana ke Youtube p dalo bdia bdia..

↓

Story Telling aa gya bdia se toh Vlogging bhi chalu kar sakte hai Youtube pr

JD based interview

Customer

Old

Story Telling Important hai..

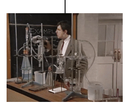
Kon si story kaha sunani hai vo bhi important hai..

Qiagen

Phama research

Paisa = Research

Scientist



10 saal IT Team

DevOps1

DevOps2

Application

3 Tier Monothic Application

Frontend react | Backend python | Database sql server

Ye 3 tier ko deploy karne ke lie my role was to design the deployment architecture and cicd pipelines...

iske baad ki kahani sabki same hogi..

Chemicals

Mixing (Ratio)

Temperature

Drug Invention

Experiment Tracking Drug	
v1	Drug Formula, Temperature, Stable, Testing,
v2	Drug Formula, Temperature, Stable, Testing,
v3	Drug Formula, Temperature, Stable, Testing,

Baith ke sochna pdega, Dimag ko tweek karke use case pr plan karna pdega... sabko khud karna pdega

Mehnat karne se hi hoga

Frontend | Backend | Database

Trunk Based Strategy

dev pipeline | prod pipeline

main

PR

PR Pipeline

feature

experiments-tracking-infra

environments: dev, prod

modules: resource_group, virtual machine, application_gateway

pipelines: dev.yaml, prod.yaml

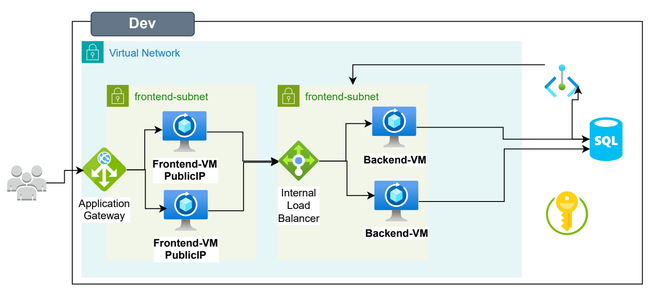
Terraform, Azure, Pipeline

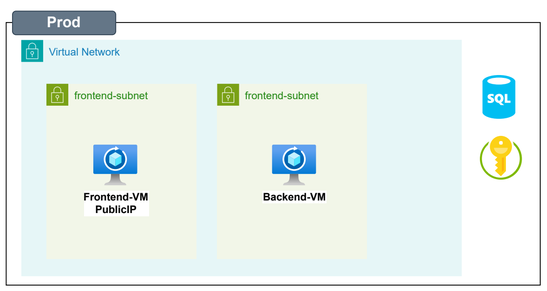
Customised as per application requirement

Stage + Gate + Round Table

Gareeb Aadmi: Decoration Consultant — Package 1, Package 2, Package 3

Ameer Aadmi → Decoration Consultant — Customised Package





I have created a deployment architecture where Application Gateway is the Entry Point of our application, there we have enable WAF and SSL Termination.

Behind the application Gateway, We had 2 virtual Machines which were running the frontend of our application

Behind the Frontend VM, we had a Internal Loadbalancer which was loadbalancing traffic between the 2 Backend Vms.

Behind the backend VMS, we had the SQL Server DB running as a PAAS Service in our azure resource group and it was connected to the VNet using Private Endpoints.

For Secret Management, We were using azure keyvault

and To deploy this complete architecture on azure cloud, we were using terraform.

So, For terrform code, we were having a git repository inside that we were having 3 folders environment, modules and pipelines.

Inside modules folder we have created all the child modules of rg, vnet, subnet, vm, applicaton gateway, lb, kv.

In the environments folder, we were having three folders dev,qa ,and prod. Inside each environment folder, the required modules for that environment were called and provider/backend config was also present

For this repository, We have implemented Trunk based branching strategy

Where we used to have all the code in main branch, all deployments happen from the main branch.

Any change that need to do to any of the environment, a feature branch was created and PR was raised.

As per our branching strategy, We have designed CICD pipelines.

such that whenever there is a PR, A PR pipeline runs and it runs tflint, tfsec, tvvalidate, tffmt check, tf plan

then the PR will be approved by alleast 2 reviewers and once the code me merged to main

then we have diferent piplenes for different environment, Based on the change, Like if the change is in dev folder in the main branch then dev pipeline will run.

In our infra pipeline, We majorly have the following stages

Terraform Init, Tfsec, tflint, tf validate, tf fmt check, tf plan, manual approval and tfapply

All our pipelines runs on self hosted agent but those agents and managed by some other team in our company.

To be Continued...

Also, my role is to design the multi environment CICD pipeline to deploy the application.

Developers have pushed their code to Github repo.

We were having 1 repo of frontend code and 1 repo for backend code.

From there, we need to build and deploy the application on the landing zone that we have deployed on azure.

For that first before designing the pipeline, we desgined the branchiing strategy for Code repositories.

For application repos, we implemented Git Flow branching strategy where we had a develop branch which was configured to be deployed on dev environment.
Adn whatever changes that comes to develop branch need to come via a feature branch.

Once the develop branch have 1 certain amount of featured created, then a release branch is created from that develop branch.

That release branch will be deployed to qa landing zone and once the testing team gives a green signal then that same branch will be deployed to prod landing zone.

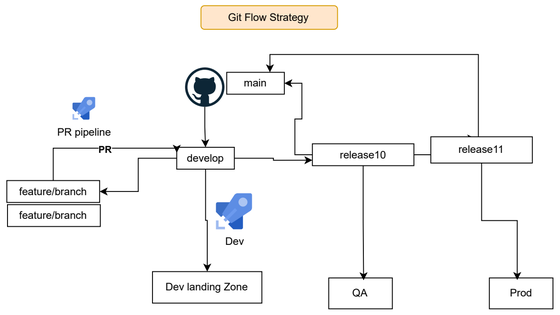
Whatever code is runninng in prod environment that code is always kept in main branch as a stable version.

As per the git flow branching stragegy, We have created a PR Pipeline that does Linting, Secret Scanning(Git Leaks), SCA scans(retire.js), SAST Scans(SonarQube), build and once the PR pipeline is successful and its reviewer by 2 approvers that code is merged to develop

As soon as code is merged to develop, dev-ui-pipleine, dev-api-pipeline runs and deploy the code on the Dev Landing Zone.

Post this, Once release branches are created, There are qa and prod pipelines that does the deployment on the qa and prod accordingly.

So like this we are managing Multi Environment Pipeline and deployment architecture for our Project.

Currently, We are working on Monothic application thats why we have not worked on Kubernetes.

Git Flow Strategy